THE BASE-CHANGE CONDITION FOR A SINGLE COMPLEX IS INHERITED BY TENSOR, INTERNAL HOM AGAINST PERFECT OBJECTS, AND PSEUDO-COHERENT INTERNAL HOM

THE STACKS PROJECT AUTHORS

Editorial note. The same-chapter quasi-coherence inputs are NOT carded (cross-line-claimed labels) and are declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 6876 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Quasi-coherence of the complexes involved. Cohomology, lemmas pseudo-coherent-pullback and perfect-pullback (cross-chapter, named); same-chapter quasi-coherence lemmas (NOT carded, declared external). The displayed maps live in D_QCoh(O_X'), making the base-change condition meaningful.
- Transitive property of the tensor product. More on Algebra, lemma triple-tensor-product (cross-chapter, named). Computes source and target of the base-change map in case (1).
- Preparation of the fancy base-change map. Cohomology, remark prepare-fancy-base-change (cross-chapter, named). First arrow of the composition used in cases (2) and (3).

Source reference. Original reference 09UA: focused-seeds/geometry/S9-perf2-single-complex-base-change-condition-inherited/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09U7: focused-seeds/geometry/S9-perf2-single-complex-base-change-condition-inherited/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08I3: focused-seeds/geometry/S9-perf2-single-complex-base-change-condition-inherited/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original equation 0DJ7: sources/stacks-perfect/pinned-original-source.tex, source lines 6644–6644. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A6H: sources/stacks-perfect/pinned-original-source.tex, source lines 2657–2657. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08DW: sources/stacks-perfect/pinned-original-source.tex, source lines 581–581. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08DX: sources/stacks-perfect/pinned-original-source.tex, source lines 620–620. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DJ8: sources/stacks-perfect/pinned-original-source.tex, source lines 6654–6654. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BYJ: focused-seeds/geometry/S9-perf2-single-complex-base-change-condition-inherited/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram of schemes. Let $K \in D_{QCoh}(\mathcal{O}_X)$ and let $L(g')^*K \to K'$ be a map in $D_{QCoh}(\mathcal{O}_{X'})$. If the equivalent conditions of Lemma 0DJ8 hold, then

1. *for $E \in D_{QCoh}(\mathcal{O}_X)$ the equivalent conditions of Lemma 0DJ8 hold for $L(g')^*(E \otimes^{\mathbf{L}} K) \to L(g')^*E \otimes^{\mathbf{L}} K'$,*
2. *if E in $D(\mathcal{O}_X)$ is perfect the equivalent conditions of Lemma 0DJ8 hold for $L(g')^*R\mathcal{H}om(E, K) \to R\mathcal{H}om(L(g')^*E, K')$, and*
3. *if K is bounded below and E in $D(\mathcal{O}_X)$ pseudo-coherent the equivalent conditions of Lemma 0DJ8 hold for $L(g')^*R\mathcal{H}om(E, K) \to R\mathcal{H}om(L(g')^*E, K')$.*

Proof. The statement makes sense as the complexes involved have quasi-coherent cohomology sheaves by Lemmas 08DW, 08DX, and 0A6H and Cohomology, Lemmas 09U7 and 09UA. Having said this, we can check the maps (0DJ7) are isomorphisms in case (1) by computing the source and target of (0DJ7) using the transitive property of tensor product, see More on Algebra, Lemma 0BYJ. The map in (2) and (3) is the composition

$$L(g')^*R\mathcal{H}om(E, K) \to R\mathcal{H}om(L(g')^*E, L(g')^*K) \to R\mathcal{H}om(L(g')^*E, K')$$

where the first arrow is Cohomology, Remark 08I3 and the second arrow comes from the given map $L(g')^*K \to K'$. To prove the maps (0DJ7) are isomorphisms one represents E_x by a bounded complex of finite projective $\mathcal{O}_{X.x}$-modules in case (2) or by a bounded above complex of finite free modules in case (3) and computes the source and target of the arrow. Some details omitted. □